

TECHNOLOGIES

REPORT

- **PYTHON 3:** The choice of this programming language was made due to its ease of use and its importance within Cybersecurity. For this reason, it was selected as the main technology for the project.
- **OOP (OBJECT-ORIENTED PROGRAMMING):** OOP was chosen to better organise the code and make the maintenance and expansion of the project easier.
- **FUNCTIONS:** Functions were implemented to avoid code repetition and make the program more organised and less repetitive.
- **CONDITIONAL STRUCTURES:** Used to control menus, games, and simulator navigation, allowing the program to make decisions.
- **LOOP STRUCTURES:** Commands that execute tasks repeatedly. They were used to keep menus active and repeat specific actions until the user chooses to exit.
- **LISTS:** Data structures capable of storing multiple elements, allowing information to be organised in a simple and efficient way.
- **DICTIONARIES:** Data structures that store information in key-value pairs, making it easier to organise and retrieve data.
- **EXCEPTION HANDLING:** A mechanism used to deal with errors during program execution. It helps prevent unexpected crashes caused by invalid user input.
- **DATE AND TIME HANDLING:** A set of resources used to work with dates and times. It was implemented to provide calendar and clock-related features.
- **COMMAND-LINE INTERFACE (CLI):** A text-based interface used through the terminal. It was chosen because it matched the knowledge acquired during development and was suitable for a study project.
- **CONCLUSION:**

The Linux Simulator was developed using Python and several fundamental programming concepts, allowing practical application of Object-Oriented Programming, functions, conditional statements, loop structures, data management, and error handling. The project contributed significantly to the development of programming skills and software organisation.